# Traction Force Microscopy for viscoelastic substrates

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## TFM and Viscoelasticity

Traction Force Microscopy (TFM) has emerged as a key tool for analyzing forces exerted by cells on substrates. From the measured displacement field of a deformable substrate, the traction field is recovered by solving an inverse of the elasticity problem [1]. Two and three-dimensional implementations are founded on the application of the Fourier transform [3], which requires the assumption of a linear elastic substrate [2]. However, it has been shown that substrates may have an important viscous contribution, and that cells respond also according to viscous forces [4]. Notably, viscous matrix viscoelasticity may promote cancer progression [5], and thus deserves to be considered.

In this work we present a methodology for solving the inverse problem in viscoelasticity, which resorts to Fourier and Laplace transforms of the special and time domains, respectively [6]. We show that in myocytes, the measured tractions may vary substantially when viscous rheology is incorporated.

## Methods

We consider a substrate in a semi-infinite domain D={(x,y,z), 0<z<h}, under mechanical equilibrium,

$$\nabla\cdot\boldsymbol{\sigma}=0$$

with $\boldsymbol{\sigma}$ Cauchy stress-tensor, and impose zero displacements at the bottom, $\boldsymbol{u}(x,y,0,t)=0$, and unknown applied traction field at the top surface:

$$\boldsymbol{t}=\boldsymbol{\sigma}\boldsymbol{n}, \text{ at } z=h$$

After assuming a linear viscoelastic material model relating $\boldsymbol{\sigma}$ and the displacement and velocity gradients, we simplify the integro-differential equations (IDE) by applying the Laplace transform to u in the time variable, and the apply Fourier transform in the space variable, resulting in a displacement field $\hat{\boldsymbol{u}}(\boldsymbol{k},z,s)$, with $\boldsymbol{k}$ the Fourier wave vector. The IDE turns into the following linear expression

$$\hat{\boldsymbol{t}}=\hat{\mathbf{A}}\hat{\mathbf{u}}$$

with Â=Â(k,s) the viscoelastic Green's function. See [6] fr further details. The tractions are transformed back to space and time variables. Numerical Laplace transform can be incorporate for more complex material models.

## Results

We compare our viscoelastic TFM (veTFM) with classical elastic versions (eTFM). Figure 1 shows the case of an epithelial cells MCF10A, where differences can reach 15% on the recovered traction field between a purely elastic and viscoelastic substrate.

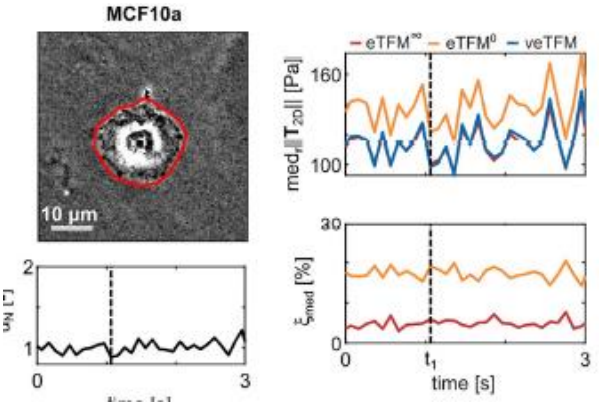



*Figure 1: Experimental application of veTFM in MCF10A epithelial cell. Top: Representative phase-contrast image. Scale bars, 10 μm. Bottom: Normalized time evolution of the displacement magnitude $u_N$. Right: Top: Median elastic and viscoelastic traction magnitudes. Bottom: Median relative error ξmed against eTFM0 (orange) or eTFM∞(red).*

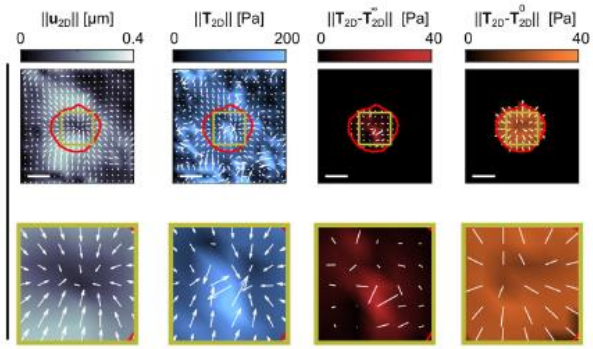


*Figure 2: Displacements, tractions, and difference wrt short and long-term elastic solutions. Bottom: Magnified views of yellow boxes.*

## Conclusions

The work presents a novel method of TFM for viscoelastic TFM without resorting to FEM techniques.

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